

ASSIGNMENT PROBLEM

PROBLEM :

The problem can be stated in a mathematical model. Below the problem is stated for a case where there are 3 jobs and 3 machines. The cost of producing job 1 on machine 1 USD but 2 USD when producing it on machine 2. Job 2 costs 2 USD on machine 1 and 3 USD on machine 2. Job 3 costs 5 USD on machine 1 and 1 USD on machine 2. Machine 3 can perform job 1 for 2 USD and jobs 2 and 3 for 3 USD respectively.

AIM :

When you have n jobs that need to be manufactured during the upcoming shift (in a manufacturing plant) and you have m machines to produce these tasks, then you want to assign the jobs to machines in an optimal way. In this say you might want to reduce the manufacturing costs incurred, hence you want to find the cost optimal production plan. The constraint, in this example, is that each machine can only fulfill one job during the upcoming shift. All jobs needs to be scheduled, i.e, assigned to a machine.

DECISION VARIABLES, OBJECTIVE AND CONSTRAINTS :

$$\min 1x_{11} + 2x_{12} + 3x_{13} + 2x_{21} + 3x_{22} + 3x_{23} + 5x_{31} + 1x_{32} + 3x_{33}$$

s.t.

$$x_{11} + x_{12} + x_{13} = 1$$

$$x_{21} + x_{22} + x_{23} = 1$$

$$x_{31} + x_{32} + x_{33} = 1$$

$$x \in \mathbb{N}$$

PROGRAM AND OUTPUT :

```
library(lpSolve)
```

```
m=rbind(c(1,2,3),c(2,3,3),c(5,1,3))
```

```
sol=lp.assign(cost.mat = m,direction = "min")
```

```
sol$solution
```

```
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    0    1
## [3,]    0    1    0
```

RESULT :

The table above shows the optimum combination of job assigned to the machine while satisfying the constraints. Job 1 must be assigned to machine 1, Job 2 must be assigned to machine 3, Job 3 must be assigned to machine 2 respectively.

QUADRATIC PROGRAMMING

PROBLEM :

$$f(x) = \frac{1}{2}x_1^2 + x_2^2 - x_1x_2 - 2x_1 - 6x_2$$

subject to the constraints

$$x_1 + x_2 \leq 2$$

$$-x_1 + 2x_2 \leq 2$$

$$2x_1 + x_2 \leq 3.$$

AIM :

To find the minimal solution of above quadratic problem using R.

PROGRAM AND OUTPUT :

```
library(quadprog)
D=matrix(c(1, -1, -1, 2), 2, 2, byrow=T)
d=c(2, 6)
A=rbind(c(-1, -1), c( 1, -2), c(-2, -1))
b=c(-2, -2, -3)

sol=solve.QP(D, d, t(A), c(b), meq=0)

res=sol$solution
cat("minimum solution : ", res, "\n")

## minimum solution :  0.6666667 1.333333

val=sol$value
cat("minimum value : ", val, "\n")

## minimum value :  -8.222222
```

RESULT :

The minimization occurs at $X_1 = 0.667$ and $X_2 = 1.333$, which results in a solution of -8.222 and satisfies all the constraints. There we have it. By breaking down the complex, intimidating QP problem, we were able to find a solution relatively easily.